

Functional Features included in the solution

Date	15 JUNE 2026
Team ID	XXXXXX
Project Name	Sustainable Waste Management Assistant Using Generative AI
Maximum Marks	5 Marks

Functional Features Included in the Solution:

Provide a comprehensive list of all functional features that have been successfully integrated into your final solution. This helps evaluators understand the scope of your project, the modules you have developed, and the exact capabilities your application provides to the end users.

S.No	Feature Name	Description & Purpose	Module / Layer	Status
1	User Authentication	Secure login and registration using Firebase Auth with email/Gmail options	Auth Module	Completed
2	Waste Item Classification (AI)	Classifies waste into categories (recyclable, organic, hazardous, e-waste) using Groq AI	Core Logic Service	Completed
3	Disposal Guidance	Provides instructions, hazard warnings, recycling steps, and eco-friendly suggestions	Core Logic Service	Completed
4	Collection Center Lookup	Displays nearby centers via Leaflet.js + OpenStreetMap with filters by waste type	Map Module	Completed

5	Analytics Dashboard	Visualizes scan history, recycling rate, hazardous items, and trends using Chart.js	Frontend UI	Completed
6	Scan History Records	Stores and retrieves user activity logs in Firebase Firestore	Database Layer	Completed
7	Eco-Friendly Recommendations	Suggests alternatives (e.g., reusable bottles, safe disposal practices)	Core Logic Service	Completed